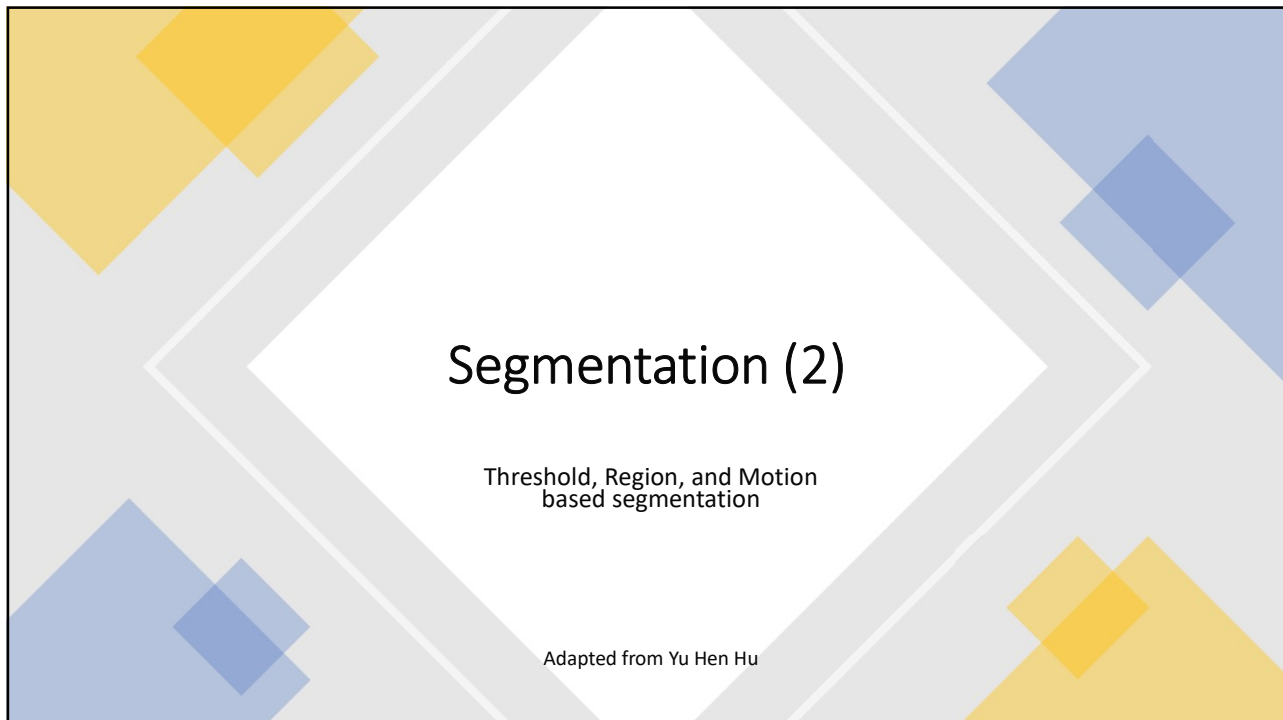
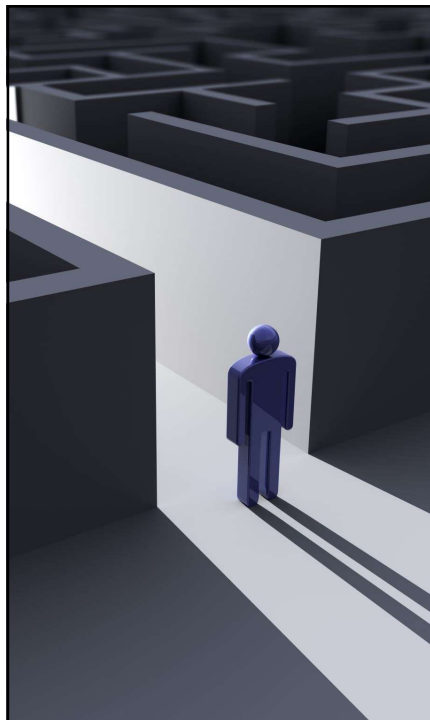




1

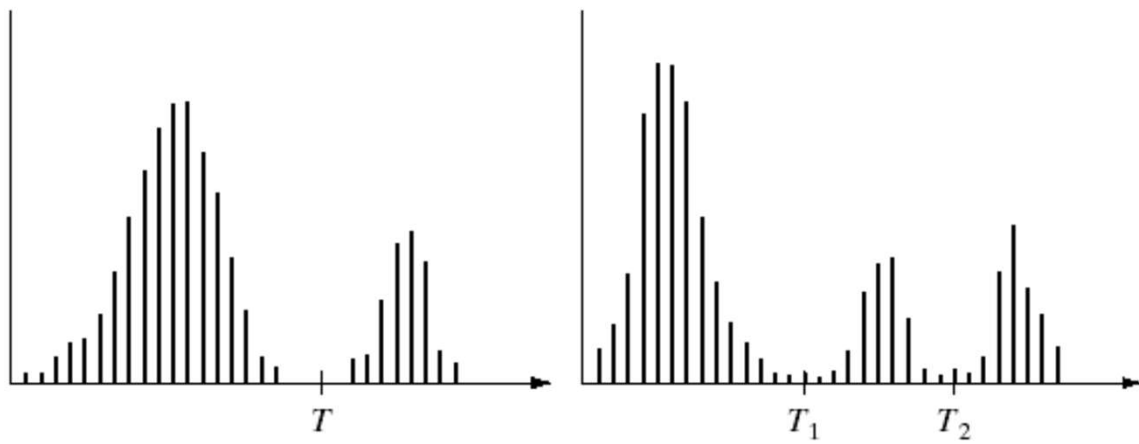


Segmentation

- Edge based segmentation
- Finding boundary between adjacent regions
- **Threshold based segmentation**
- Finding regions by grouping pixels with similar gray-values
- Region based segmentation
- Finding regions directly using growing or splitting
- Motion based segmentation
- Finding regions by comparing successive frames of a video sequence to identify regions that correspond to moving objects

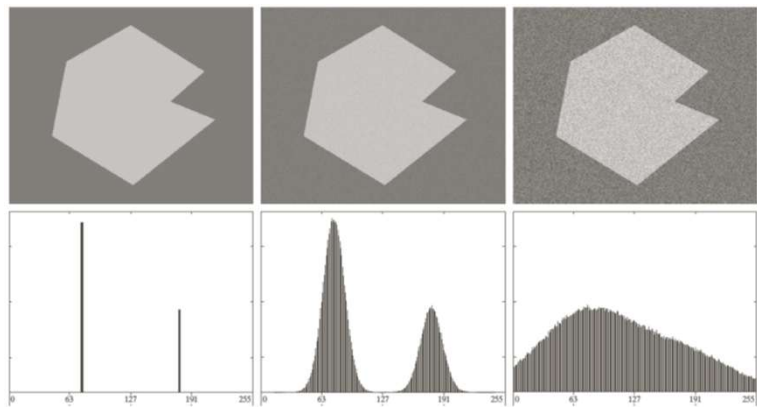
2

Threshold Segmentation



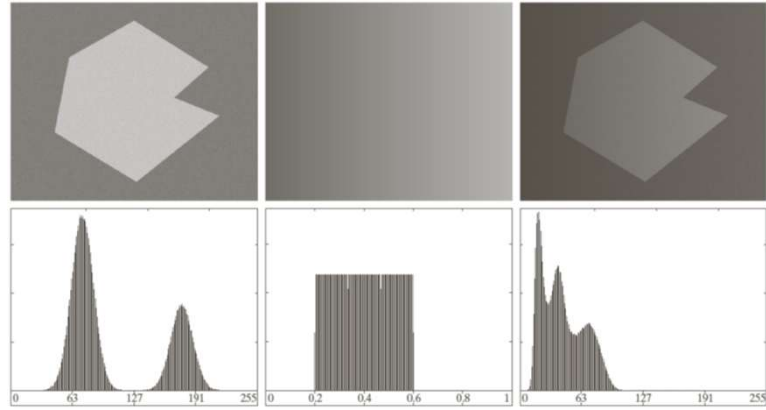
3

Effect of
Noise in
Thresholding



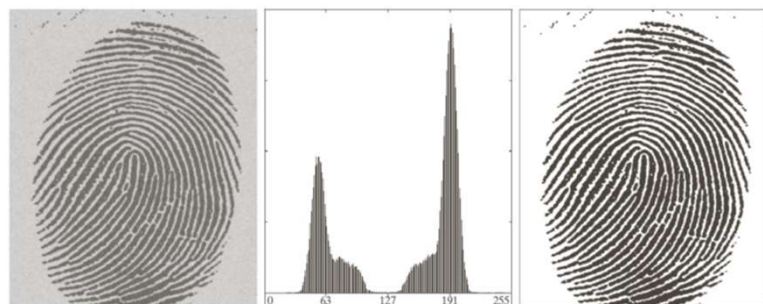
4

Effect of Illumination on Threshold



5

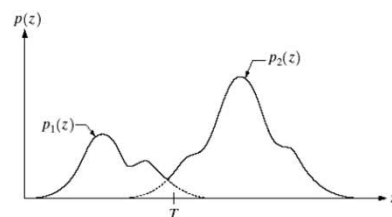
Fingerprint Example



6

Threshold: Hypothesis Testing

- Question: Does this pixel with intensity z belong to a region (edge) or not?
- Hypothesis
 - H_0 : Null. It does not
 - H_1 : Alt. It does
- Likelihood
 - $p(z|z \in H_0) = p_1(z)$
 - $p(z|z \in H_1) = p_2(z)$
- Prior
 - $P_1 = p(z \in H_0)$,
 - $P_2 = p(z \in H_1)$
- Maximum likelihood:
 - Pixel z belongs to a region if $p(z|H_1) > p(z|H_0)$
- Bayesian:
 - $P_2 p(z|H_1) > P_1 p(z|H_0)$
- Sufficient statistic:
 - $z > T$



7

Uni-model Gaussian Example

- Given

$$p_i(z) = \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left(-\frac{(z - \mu_i)^2}{2\sigma_i^2}\right)$$
 - Set

$$P_1 p_1(T) = P_2 p_2(T)$$
 and solve for T .
 - Take log on both sides and simplify to

$$AT^2 + BT + C = 0$$
- $$A = \sigma_1^2 - \sigma_2^2$$
- $$B = 2(\mu_1\sigma_2^2 - \mu_2\sigma_1^2)$$
- $$C = \sigma_1^2\mu_2^2 - \sigma_2^2\mu_1^2 + 2\sigma_1^2\sigma_2^2 \ln\left(\frac{\sigma_2 P_1}{\sigma_1 P_2}\right)$$
- If $\sigma_1 = \sigma_2 = \sigma$, then
- $$T = \frac{\mu_1 + \mu_2}{2} + \frac{\sigma^2}{\mu_1 - \mu_2} \cdot \ln\left(\frac{P_2}{P_1}\right)$$

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Clustering Problem Statement

- Given a set of vectors $\{x_k; 1 \leq k \leq K\}$, find a set of M clustering centers $\{w(i); 1 \leq i \leq c\}$ such that each x_k is assigned to a cluster, say, $w(i^*)$, according to a distance (distortion, similarity) measure $d(x_k, w(i))$ such that the average distortion

$$D = \frac{1}{K} \sum_{i=1}^c \sum_{k=1}^K I(x_k, i) d(x_k, w(i))$$

is minimized.

- $I(x_k, i) = 1$ if x is assigned to cluster i with cluster center $w(i)$; and $= 0$ otherwise -- indicator function.

9

k-means Clustering Algorithm

Initialization: Initial cluster center $w(i); 1 \leq i \leq c, D(-1) = 0, I(x_k, i) = 0, 1 \leq i \leq c, 1 \leq k \leq K;$

Repeat

(A) Assign cluster membership (Expectation step)

Evaluate $d(x_k, w(i)); 1 \leq i \leq c, 1 \leq k \leq K$

$I(x_k, i) = 1$ if $d(x_k, w(i)) < d(x_k, w(j)), j \neq i;$

$= 0;$ otherwise. $1 \leq k \leq K;$

(B) Evaluate distortion D :

$$D(\text{iter}) = \sum_{k=1}^N I(x_k, i) d(x_k, w(i)) \quad 1 \leq k \leq K$$

(C) Update code words according to new assignment (Maximization)

$$w(i) = \sum_{k=1}^N I(x_k, i) x_k, \quad N_i = \sum_{k=1}^N I(x_k, i), \quad 1 \leq i \leq c$$

(D) Check for convergence

if $1 - D(\text{Iter}-1)/D(\text{Iter}) < \varepsilon$, then convergent = TRUE,

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Numerical Example

$x = \{-1, -2, 0, 2, 3, 4\}$,

$W = \{2.1, 2.3\}$

- **Assign membership**

2.1: $\{-1, -2, 0, 2\}$

2.3: $\{3, 4\}$

- **Distortion**

$$D = (-1 - 2.1)^2 + (-2 - 2.1)^2 + (0 - 2.1)^2 + (2 - 2.1)^2 + (3 - 2.3)^2 + (4 - 2.3)^2$$

- **3. Update W to minimize distortion**

$$W_1 = (-1 - 2 + 0 + 2)/4 = -.25$$

$$W_2 = (3 + 4)/2 = 3.5$$

- **4. Reassign membership**

-.25: $\{-1, -2, 0\}$

3.5: $\{2, 3, 4\}$

- **5. Update W:**

$$w_1 = (-1 - 2 + 0)/3 = -1$$

$$w_2 = (2 + 3 + 4)/3 = 3.$$

- **Converged.**

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Otsu's Optimal Threshold

- Assume there are L levels of intensity, and n_i is # of pixels with intensity level i . then,

$$MN = \sum_{i=1}^L n_i \quad \text{or} \quad 1 = \sum_{i=1}^L \frac{n_i}{MN} = \sum_{i=1}^L p_i \quad 0 \leq p \leq 1.$$

- Assume the intensity values are divided into 2 groups $\{0, 1, \dots, k\}$ and $\{k+1, \dots, L-1\}$. The probabilities a pixel is in each group are:

$$P_1(k) = \sum_{i=1}^k p_i \quad \text{and} \quad P_2(k) = \sum_{i=k+1}^{L-1} p_i. \quad \text{Note that } P_1(k) + P_2(k) = 1$$

- The mean values of each group and entire images are:

$$m_1(k) = \frac{1}{P_1(k)} \sum_{i=0}^k i \cdot p_i; \quad m_2(k) = \frac{1}{P_2(k)} \sum_{i=k+1}^{L-1} i \cdot p_i; \quad m_G = \sum_{i=0}^{L-1} i \cdot p_i$$

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Otsu's Optimal Threshold (2)

- Define "between class variance"

$$\sigma_B^2(k) = P_1(k)[m_1(k) - m_G]^2 + P_2(k)[m_2(k) - m_G]^2$$

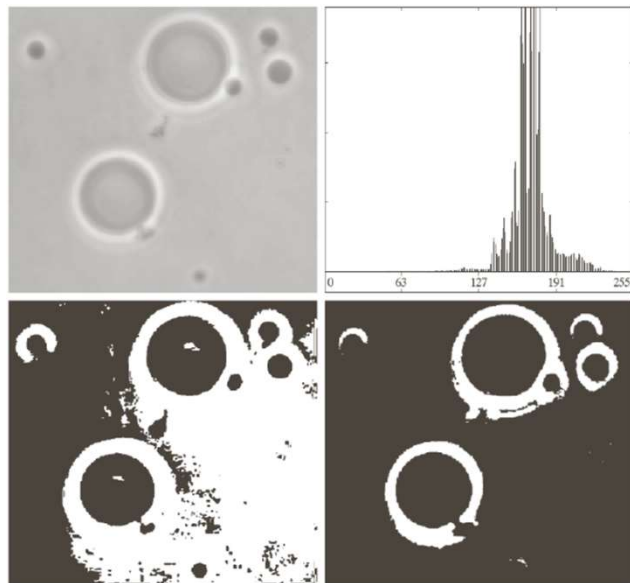
- Clustering result is "optimal" if a threshold $T(k^*)$ is selected such that

$$\sigma_B^2(k^*) = \min_{1 \leq k \leq L-1} \sigma_B^2(k)$$

- Otsu's algorithm: compute $\sigma_B^2(k)$ for $k = 1, 2, \dots$ till a minimum is found.

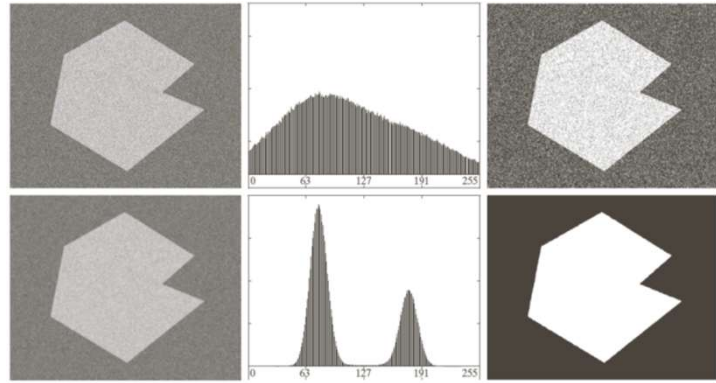
13

Threshold
Methods
Comparison



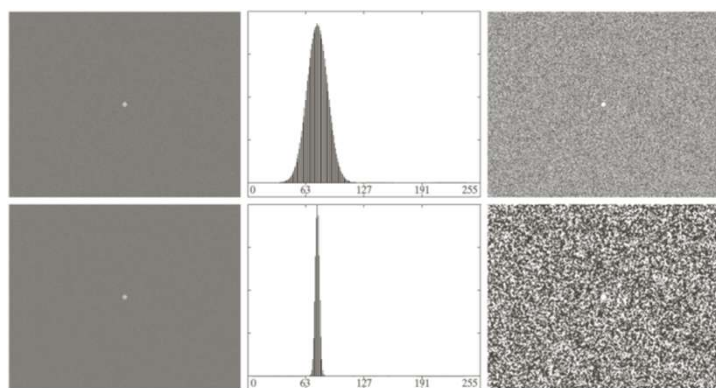
14

Using
Smoothing to
Improve
Threshold



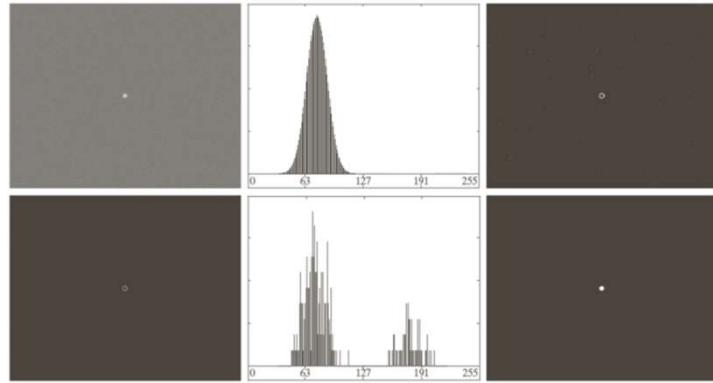
15

Failure due
to Uneven
Priors



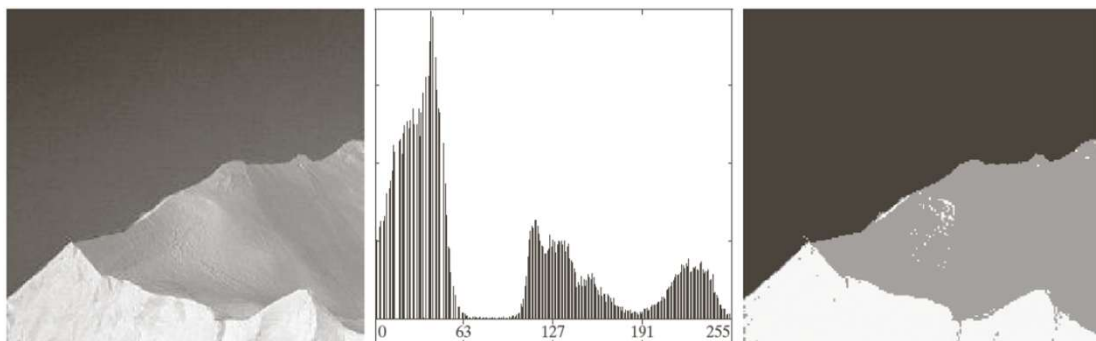
16

Using Edges
to Improve
Threshold



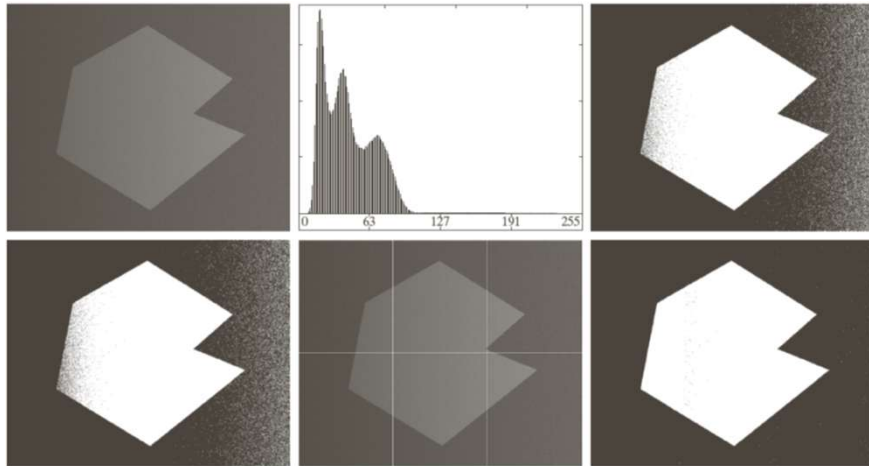
17

Multi-threshold



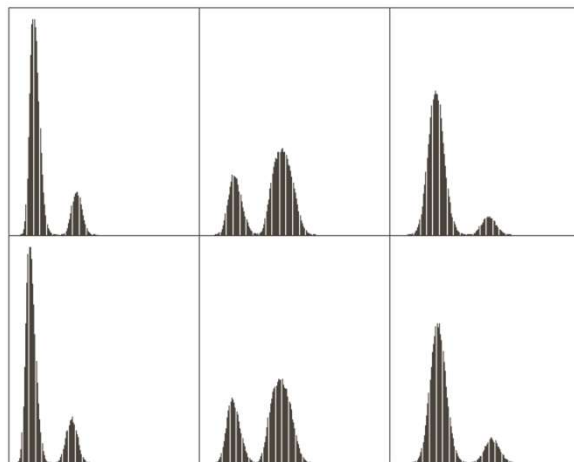
18

Local Threshold



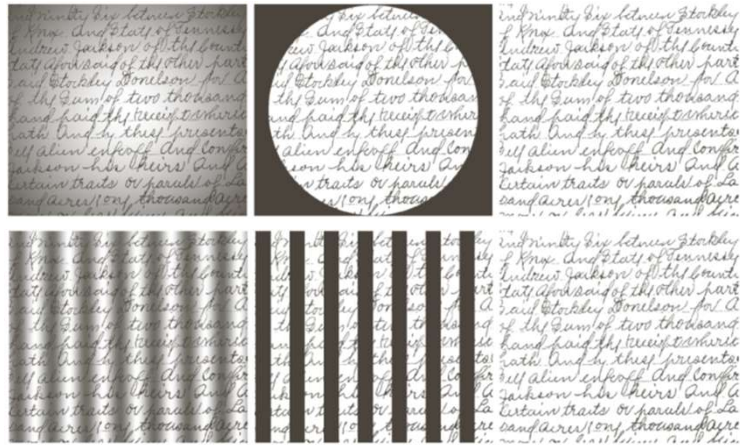
19

Local Threshold



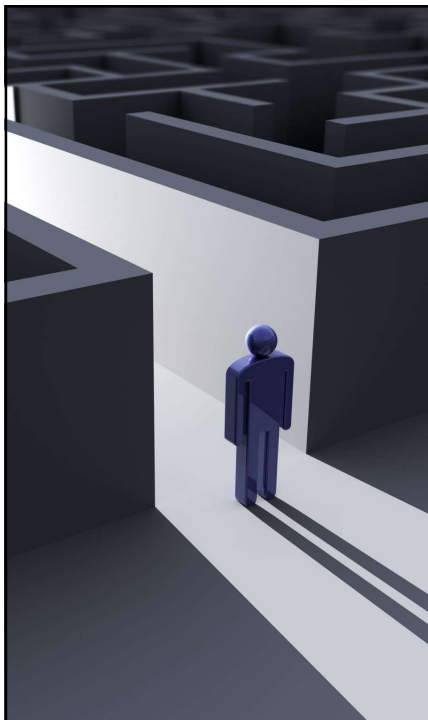
20

Local Threshold with Sliding Window



Image, results of Otsu, results of moving averages

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Segmentation

- Edge based segmentation
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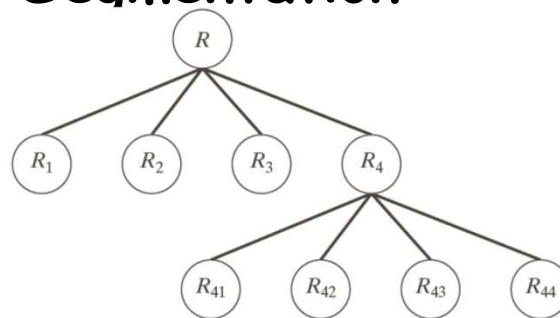
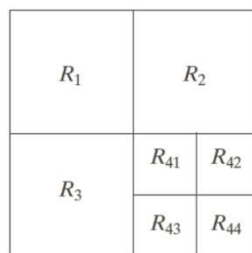
22

Region-based Segmentation Methods

- Region-Growing
 - Start with seed points
 - Grow around current seed regions by attaching pixels of the same property to the growing sub-regions.
- Region splitting and Merging
 - Apply the predicate to a sub-region. If it is true, stop splitting. Else, split the region.
 - Quad-tree is one way of splitting.
 - Neighboring sub-regions with the same predicates can be merged.

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Region-based Segmentation

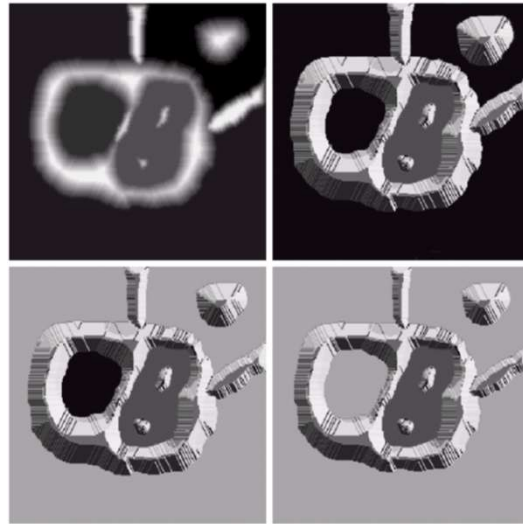


- A segmentation is the partition of an image R into sub-regions $\{R_i\}$ such that each R_i is a connected region, and

$$\bigcup_{i=1}^n R_i = R; \quad R_i \cap R_j = \emptyset$$
- A region can be defined by a **predicate** P such that $P(R_i) = \text{TRUE}$ if all pixels within the region satisfy a specific property.
 $P(R_i \cap R_j) = \text{FALSE}$ for $i \neq j$.

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Watersheds Segmentation

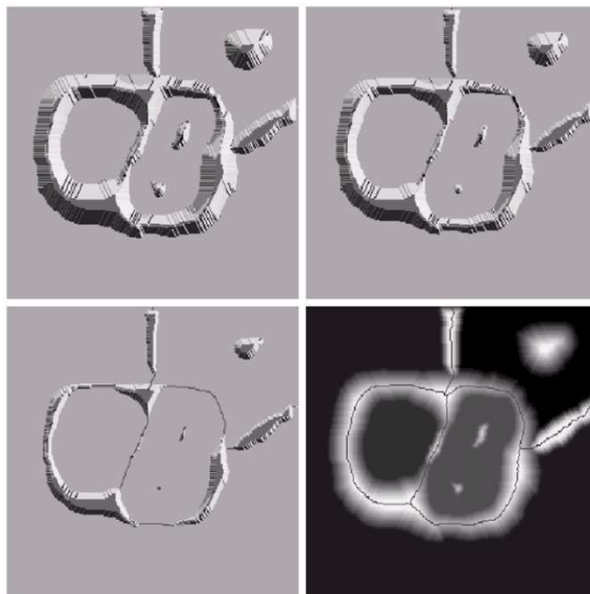


a	b
c	d

FIGURE 10.54

(a) Original image.
(b) Topographic view. (c)–(d) Two stages of flooding.

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e	f
g	h

FIGURE 10.54
(Continued)

(e) Result of further flooding.
(f) Beginning of merging of water from two catchment basins (a short dam was built between them). (g) Longer dams. (h) Final watershed (segmentation) lines.

(Courtesy of Dr. S. Beucher, CMM/Ecole des Mines de Paris.)

Watersheds Segmentation

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Watersheds Segmentation

<http://cmm.enscm.fr/~beuc>
her/wtshed.html

A morphological region growing approach.

Seed points: local minima points

Growing method: Dilation

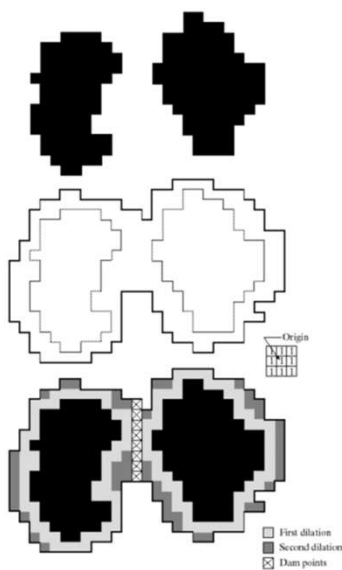
Predicates Similar gradient values

Sub-region boundary Dam building

To avoid over-segmentation Use markers

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Dam Building



- **FIGURE 10.55** (a) Two partially flooded catchment basins at stage $n-1$ of flooding. (b) Flooding at stage n , showing that water has spilled between basins (for clarity, water is shown in white rather than black). (c) Structuring element used for dilation. (d) Result of dilation and dam construction.

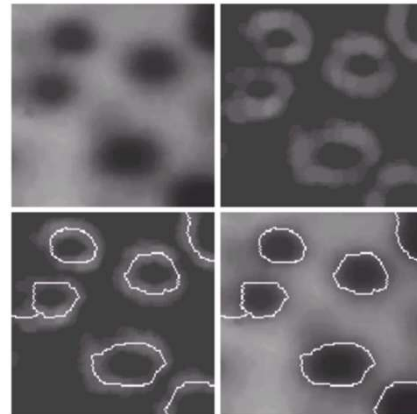
28

Watershed Segmentation Example

a b
c d

FIGURE 10.56

(a) Image of blobs.
(b) Image gradient.
(c) Watershed lines.
(d) Watershed lines superimposed on original image.
(Courtesy of Dr. S. Beucher, CMM/Ecole des Mines de Paris.)

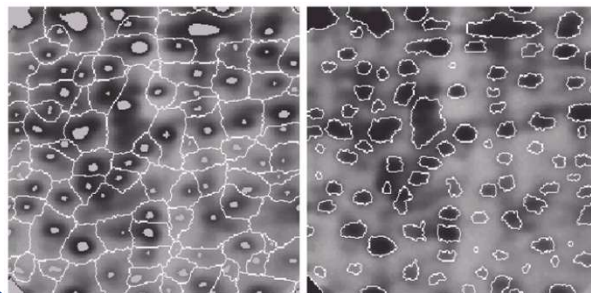
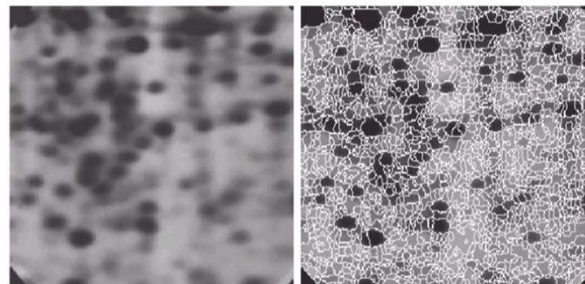


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Over-Segmentation and Use of Marker

a b

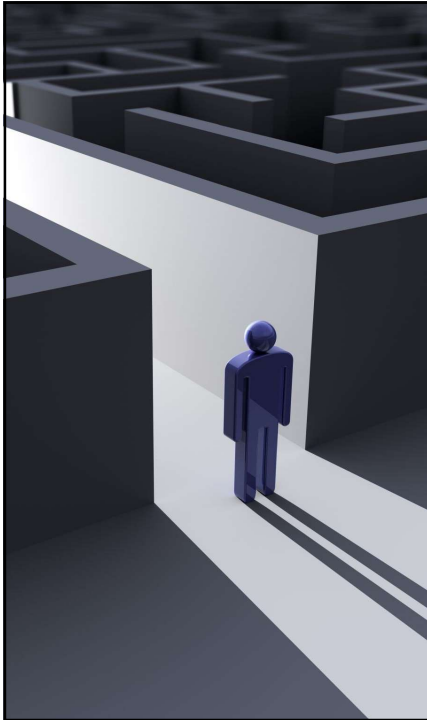
FIGURE 10.57
(a) Electrophoresis image. (b) Result of applying the watershed segmentation algorithm to the gradient image. Oversegmentation is evident.
(Courtesy of Dr. S. Beucher, CMM/Ecole des Mines de Paris.)



a b

FIGURE 10.48
(a) Image showing internal markers (light gray regions) and external markers (watershed lines). (b) Result of segmentation. Note the improvement over Fig. 10.47(b).
(Courtesy of Dr. S. Beucher, CMM/Ecole des Mines de Paris.)

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Segmentation

- Edge based segmentation
 - Finding boundary between adjacent regions
- Threshold based segmentation
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- Region based segmentation
 - Finding regions directly using growing or splitting
- **Motion based segmentation**
 - Finding regions by comparing successive frames of a video sequence to identify regions that correspond to moving objects

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Motion-Based Segmentation

Use motion as a cue to distinguish moving objects in an image sequence (video)

Assumption:

regions that moves are of interests than still backgrounds

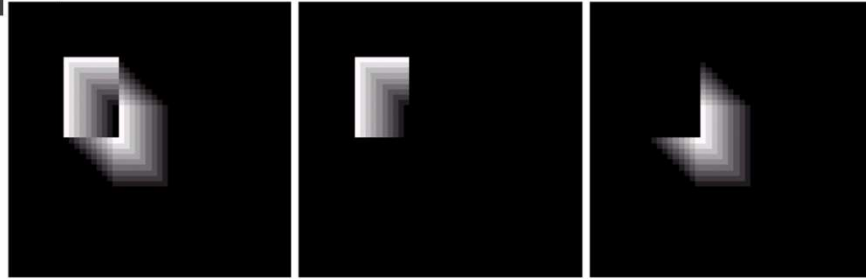
Successive image frames are taken from the same camera at the same shot.

Spatial versus frequency domain methods

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Accumulative Difference Image

- Difference Image $d_{ij}(x, y) = \begin{cases} 1 & \text{if } |f(x, y, t_i) - f(x, y, t_j)| > T \\ 0 & \text{otherwise.} \end{cases}$
- Absolute ADI: $A_k(x, y) = \sum_{i=1}^k d_{0,i}(x, y)$



a b c

FIGURE 10.59 ADIs of a rectangular object moving in a southeasterly direction. (a) Absolute ADI. (b) Positive ADI. (c) Negative ADI.

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ADI Calculations

- Absolute, positive, and negative ADIs
- $R(x, y)$: original image
- $f(x, y, k) = f(x, y, tk)$ k -th frame in a sequence of images to be compared to $R(x, y)$

$$A_k(x, y) = \begin{cases} A_{k-1}(x, y) + 1 & |R(x, y) - f(x, y, k)| > T \\ A_{k-1}(x, y) & \text{otherwise.} \end{cases}$$

$$P_k(x, y) = \begin{cases} P_{k-1}(x, y) + 1 & R(x, y) - f(x, y, k) > T \\ P_{k-1}(x, y) & \text{otherwise.} \end{cases}$$

$$N_k(x, y) = \begin{cases} N_{k-1}(x, y) + 1 & R(x, y) - f(x, y, k) < -T \\ N_{k-1}(x, y) & \text{otherwise.} \end{cases}$$

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Terimakasih